

# Shun Zhang

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🌐 [shunzh.github.io](https://shunzh.github.io)

Research Interests Reinforcement learning; foundation models; code generation and reasoning; value alignment; artificial general intelligence.

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## Experience

- Aug 2025 – Present **Senior Generative AI Engineer**, *NVIDIA*, Santa Clara, CA  
Training and evaluating **vision-language models (VLMs)** and **world action models (WAMs)**.  
Developing inference-time scaling methods for WAMs and building Isaac Sim-based evaluation environments.
- Jun 2024 – Jan 2025 **Founding Member of Technical Staff**, *Asari AI*, San Francisco, CA  
Developed a **code agent** that plans, verifies, and discovers new skills and knowledge.
- Jun 2022 – Jun 2024 **Research Scientist**, *MIT-IBM Watson AI Lab*  
Research on **reinforcement learning** and post-training of **language models**, with a focus on code generation and reinforcement learning from human feedback (RLHF).
- Aug 2020 – Jun 2022 **Postdoctoral Researcher**, *IBM Research*  
Research on **meta-reinforcement learning** and **AI for scientific discovery**.

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## Education

- Sep 2015 – Apr 2020 **Ph.D. in Computer Science and Engineering**, *University of Michigan*, Ann Arbor, MI  
Advisors: Prof. Satinder Singh and Prof. Ed Durfee.  
Research on value alignment and AI safety in reinforcement learning.
- Aug 2015 **B.S. and M.S. in Computer Science**, *University of Texas at Austin*, Austin, TX  
Undergraduate/master research advisors: Prof. Peter Stone and Prof. Dana Ballard.

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## Publications and Preprints

- 2026 **Cosmos 3: Omnimodal World Models for Physical AI**  
NVIDIA  
*arXiv* 
- 2024 **Improving Reinforcement Learning from Human Feedback with Efficient Reward Model Ensemble**  
**Shun Zhang**, Zhenfang Chen, Sunli Chen, Yikang Shen, Zhiqing Sun, Chuang Gan  
*arXiv* 
- 2024 **Graph-Transformer-based Surrogate Model for Accelerated Converter Circuit Topology Design**  
Shaoze Fan, Haoshu Lu, **Shun Zhang**, Ningyuan Cao, Xin Zhang, Jing Li  
*Design Automation Conference (DAC)* 
- 2023 **Adaptive Online Replanning with Diffusion Models**  
Siyuan Zhou, Yilun Du, **Shun Zhang**, Mengdi Xu, Yikang Shen, Wei Xiao, Dit-Yan Yeung, Chuang Gan  
*Conference on Neural Information Processing Systems (NeurIPS)* 
- 2023 **Planning with Large Language Models for Code Generation**  
**Shun Zhang**, Zhenfang Chen, Yikang Shen, Mingyu Ding, Joshua B. Tenenbaum, Chuang Gan  
*International Conference on Learning Representations (ICLR)* 
- 2023 **Hyper-Decision Transformer for Efficient Online Policy Adaptation**  
Mengdi Xu, Yuchen Lu, Yikang Shen, **Shun Zhang**, Ding Zhao, Chuang Gan  
*International Conference on Learning Representations (ICLR)* 

- 2022 **Prompting Decision Transformer for Few-shot Policy Generalization**  
Mengdi Xu, Yikang Shen, **Shun Zhang**, Yuchen Lu, Ding Zhao, Joshua B. Tenenbaum, Chuang Gan  
*International Conference on Machine Learning (ICML)* 
- 2022 **Power Converter Circuit Design Automation using Parallel Monte Carlo Tree Search**  
Shaoze Fan, **Shun Zhang**, Jianbo Liu, Ningyuan Cao, Xiaoxiao Guo, Jing Li, Xin Zhang  
*ACM Transactions on Design Automation of Electronic Systems (TODAES)* 
- 2021 **From Specification to Topology: Automatic Power Converter Design via Reinforcement Learning**  
Shaoze Fan, Ningyuan Cao, **Shun Zhang**, Jing Li, Xiaoxiao Guo, Xin Zhang  
*International Conference on Computer Aided Design (ICCAD)* 
- 2020 **Efficiently Finding Approximately-Optimal Queries for Improving Policies and Guaranteeing Safety**  
**Shun Zhang**  
*Ph.D. Dissertation* 
- 2020 **Querying to Find a Safe Policy Under Uncertain Safety Constraints in Markov Decision Processes**  
**Shun Zhang**, Edmund H. Durfee, Satinder Singh  
*AAAI Conference on Artificial Intelligence (AAAI)* 
- 2018 **Minimax-Regret Querying on Side Effects for Safe Optimality in Factored Markov Decision Processes**  
**Shun Zhang**, Edmund H. Durfee, Satinder Singh  
*International Joint Conference on Artificial Intelligence (IJCAI)* 
- 2018 **Modeling Sensory-Motor Decisions in Natural Behavior**  
Ruohan Zhang, **Shun Zhang**, Matthew H. Tong, Yuchen Cui, Constatin A. Rothkopf, Dana H. Ballard, Mary M. Hayhoe  
*PLoS Computational Biology* 
- 2017 **Approximately-Optimal Queries for Planning in Reward-Uncertain Markov Decision Processes**  
**Shun Zhang**, Edmund H. Durfee, Satinder Singh  
*International Conference on Automated Planning and Scheduling (ICAPS)* 
- 2015 **Determining Placements of Influencing Agents in a Flock**  
Katie Genter, **Shun Zhang**, Peter Stone  
*Autonomous Agents and Multiagent Systems (AAMAS)* 
- 2015 **Autonomous Intersection Management for Semi-Autonomous Vehicles**  
Tsz-Chiu Au, **Shun Zhang**, Peter Stone  
*Handbook of Transportation* 

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## Academic Services

Conference Reviewer    AAAI 2019, AISTATS 2023-24, CVPR 2023, ICML 2023-24, NeurIPS 2023-25, ICLR 2024-25.